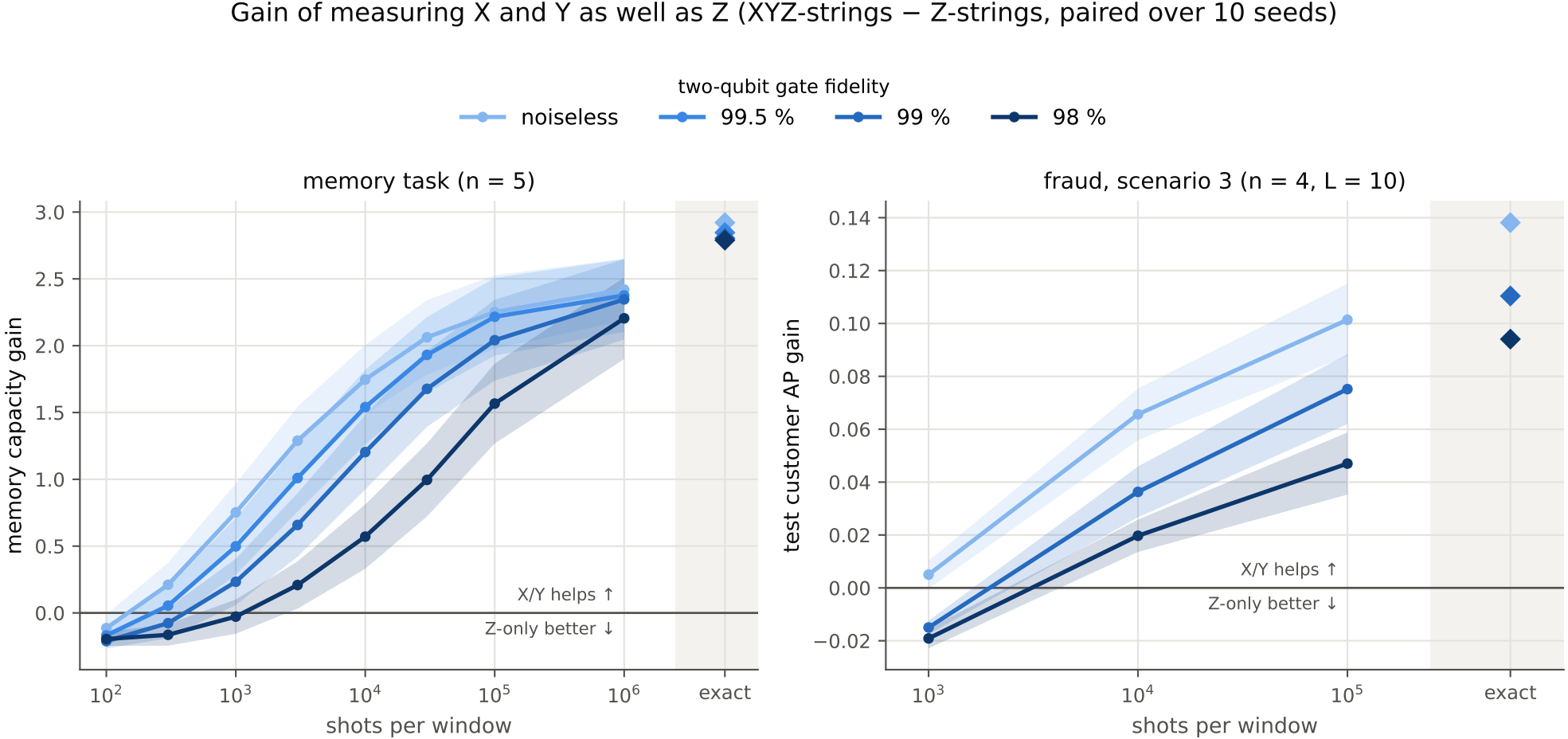
Gain of measuring X and Y as well as Z (XYZ-strings − Z-strings, paired over 10 seeds)
two-qubit gate fidelity
noiseless
99.5 %
99 %
98 %
memory task (n = 5)
memory capacity gain
3.0
2.5
2.0
1.5
1.0
0.5
0.0
X/Y helps ↑
Z-only better ↓
10^2
10^3
10^4
10^5
10^6
exact
shots per window
fraud, scenario 3 (n = 4, L = 10)
test customer AP gain
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
X/Y helps ↑
Z-only better ↓
10^3
10^4
10^5
exact
shots per window